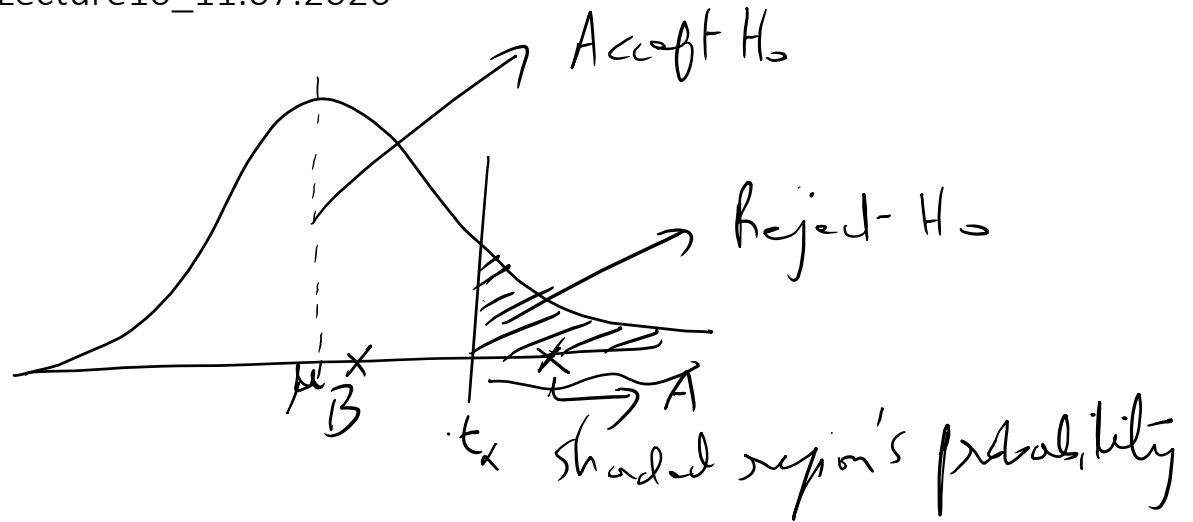




$$\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \approx t_{n-1} \left. \begin{array}{l} \text{t distribution with} \\ (n-1) \text{ degrees of freedom} \end{array} \right\} \text{Mar} = \alpha$$

In the computation of s we do

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$

$$H_0: \mu = 18580 (\mu_0)$$

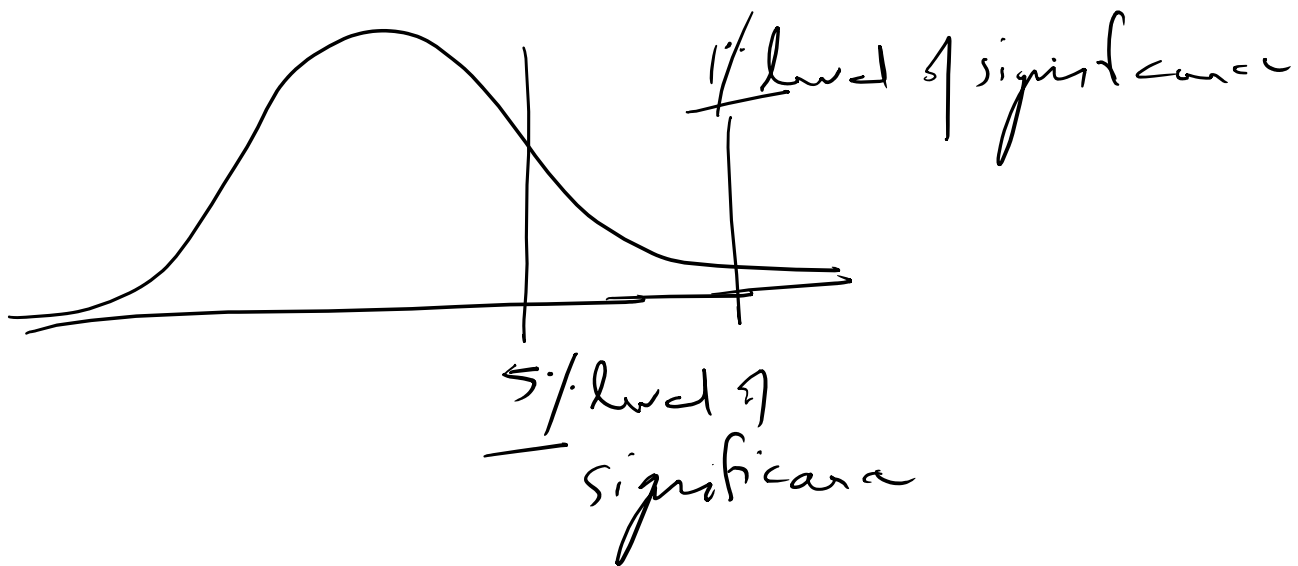
$$H_a: \mu > \mu_0$$

H_0 :
assuming
 H_0 , we can
take
 $\mu = 18580$

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$$

17352
2012

$$\frac{17352 - 18580}{\frac{2012}{\sqrt{10}}}$$



$$t_{d, n_1 + n_2 - 2} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{x}_1$ = sample mean from n_1 samples drawn from the first population

$\bar{x}_2$ = sample mean from n_2 samples drawn from the second population

s_1 = sample standard dev from Sample 1

s_2 = sample stdev from Sample 2

$$H_0: \mu_1 - \mu_2 = 0$$

Why is the denominator equal to

$$\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}$$

$$\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)$$

Since $\bar{X}_1$ and $\bar{X}_2$ are independent

$$\begin{aligned} \text{Var}(\bar{X}_1) &= \text{Var}\left(\frac{X_1 + X_2 + \dots + X_{n_1}}{n_1}\right) \\ &= \frac{1}{n_1^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_{n_1})] \end{aligned}$$

$$\text{Var}(X_1) = \text{Var}(X_2) = \dots = \text{Var}(X_{n_1}) \quad \therefore \text{ iid property}$$

We do not have σ_1^2 = population variance
 Instead we compute s_1^2 = sample variance

$$\text{Var}(\bar{X}_1) = \frac{\text{Var}(X_1)}{n_1} = \frac{s_1^2}{n_1}$$

$$\text{Var}(\bar{X}_2) = \frac{s_2^2}{n_2}$$

$$\text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2) = \frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}$$

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)}}$$

$$= \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Testing the difference between means of
paired observations

$(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$

↑
blood pressure
before tablet in
patient 1

↑
blood pressure after
tablet in patient 1

Does the tablet contribute to changing
blood pressure.

Define the differences $D_i = X_i - Y_i$

We now have a single sample

$$D_1, D_2, \dots, D_n$$

Assume $D_i \stackrel{iid}{\sim} N(\mu_D, \sigma_D^2)$

We make no assumption that X_i and Y_i are independent, we only make an assumption that differences from different pairs are independent

Let us say the tablet has no impact on
blood pressure

$$\underline{E(D_i)} = E(X_i) - E(Y_i) = \mu_X - \mu_Y = 0$$

$$\underline{H_0 : \mu_D = 0}$$

Sample statistics

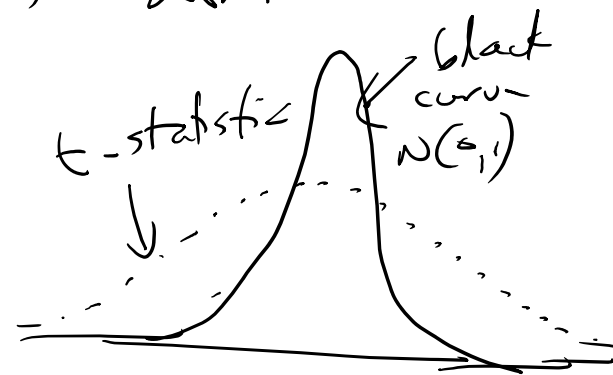
$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i$$

$$S_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2$$

$$\frac{\bar{D}}{\frac{s_D}{\sqrt{n}}} = T \rightarrow \text{a statistic } \eta$$

a student's t-dist
with $(n-1)$ degrees of
freedom

$$\frac{\bar{D}}{\frac{s_D}{\sqrt{n}}} \rightarrow N(0, 1)$$



chi-squared test for independence

- used to determine whether the categorical variables are statistically independent or not

Is a student's smoking status independent of whether they have lung disease

How do we set up the chi-squared test for independence.

Step 1: state the hypotheses:
Suppose there are two categorical variables

- Variable A with r categories
- Variable B with c categories

H_0 : Variables A and B are independent

H_a : Variables A and B are not independent

Step 2 : Collect the observed data
 The data are summarised in a
contingency table

	Coffee	Tea	Juice	Row total
Male	40	25	15	80
Female	20	35	25	80
Column total	60	60	40	160

The entries in the table are called
 observed frequencies denoted O_{ij}

Step 3: What would independence predict?
 if the variables are independent then

$$P(A=i, B=j) = P(A=i)P(B=j)$$

We estimate these probabilities from the observed marginal totals.

For cell (i, j)

$$P(A=i) = \frac{\text{Row total } i}{N}$$

$$P(B = j) = \frac{\text{Column total } j}{N}$$

$$P(A = i, B = j) \stackrel{?}{=} P(A = i) P(B = j)$$

$$\stackrel{?}{=} \left(\frac{\text{Row total } i}{N} \right) \left(\frac{\text{Column total } j}{N} \right)$$

Multiplying by the sample size N , the expected count in cell (i, j)

$$E_{ij} = \frac{(\text{Row total } i)(\text{Column total } j)}{N}$$

Example

for males who prefer coffee

$$h-w \text{ total} = 80$$

$$\text{Column total} = 60$$

$$\text{total sample} = 160$$

$$\frac{80 \times 60}{160} = \underline{30}$$

The observed count (from the contingency table) is 40

Thus, this cell contributes $\frac{(40 - 30)^2}{30}$
 $= 3.33$ to the test statistic

Step 4: Compute the chi-squared statistic

The statistic compares every observed count with its expected count

$$\chi^2_{obs} = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

if observed and expected counts are close
the contribution is small
if they differ greatly, the contribution is large

Step 5: Distribution of the statistic

If the null hypothesis is true, the sample is sufficiently large and the expected counts are not too small

$\chi^2_{obs} \sim \chi^2$ where $\uparrow \nu = (r-1)(c-1)$
 "nu" greek symbol
 is the number of degrees of freedom

We do have $r \times c$ cells for $r \times c$ cell count

however the row totals are fixed

$\hookrightarrow (r-1)$ independent pieces of information

the column totals are also fixed

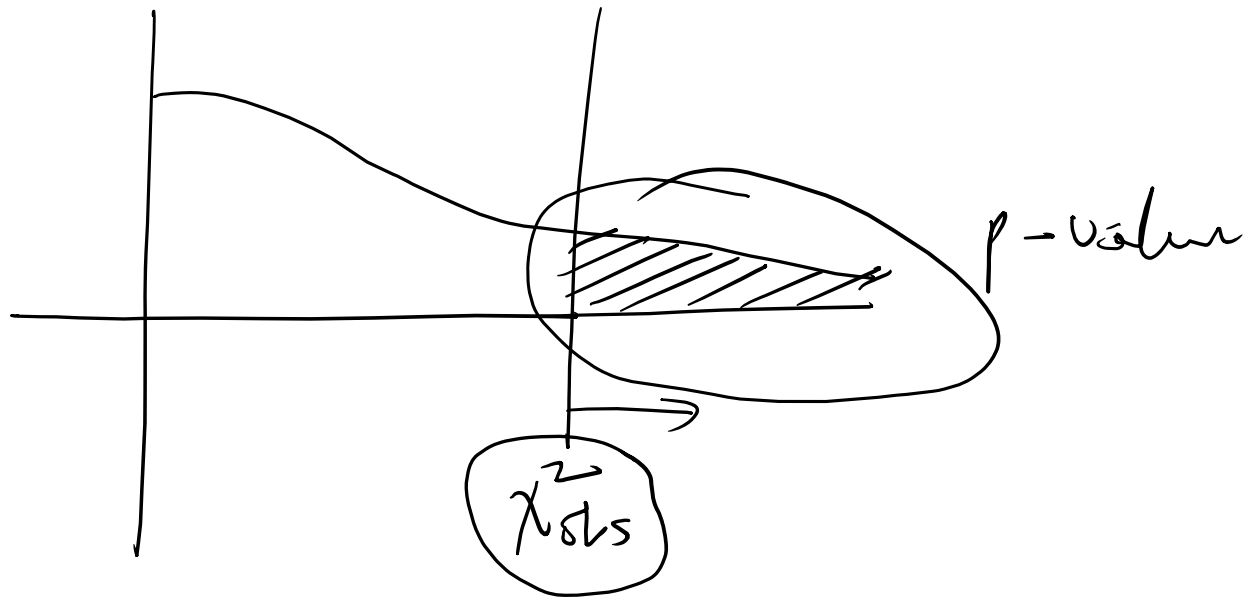
$\rightarrow (c-1)$ independent pieces of information

$$D = (r-1)(c-1)$$

Step 6: Compute the p-value
 χ^2_{obs} (coming from the table) is a
 statistic distributed according to
 chi-squared distribution with
 $\nu = (r - 1)(c - 1)$ degrees of freedom

$$P(\chi^2 \geq \chi^2_{obs}) = p\text{-value}$$

if $(p < \alpha)$ reject H_0



Why does the test statistic have this form

Each observed count fluctuates around its expected count

$$O_{ij} \approx E_{ij} + \text{random error}$$

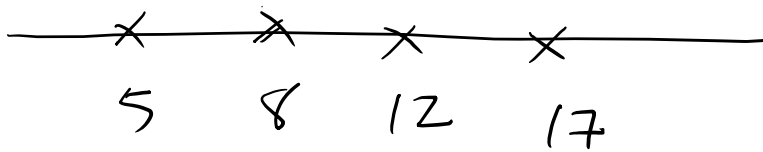
$$\text{Var}(O_{ij}) \approx E_{ij} \rightarrow \text{assumption we make.}$$

Therefore

$$\frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}} \text{ is approximately}$$

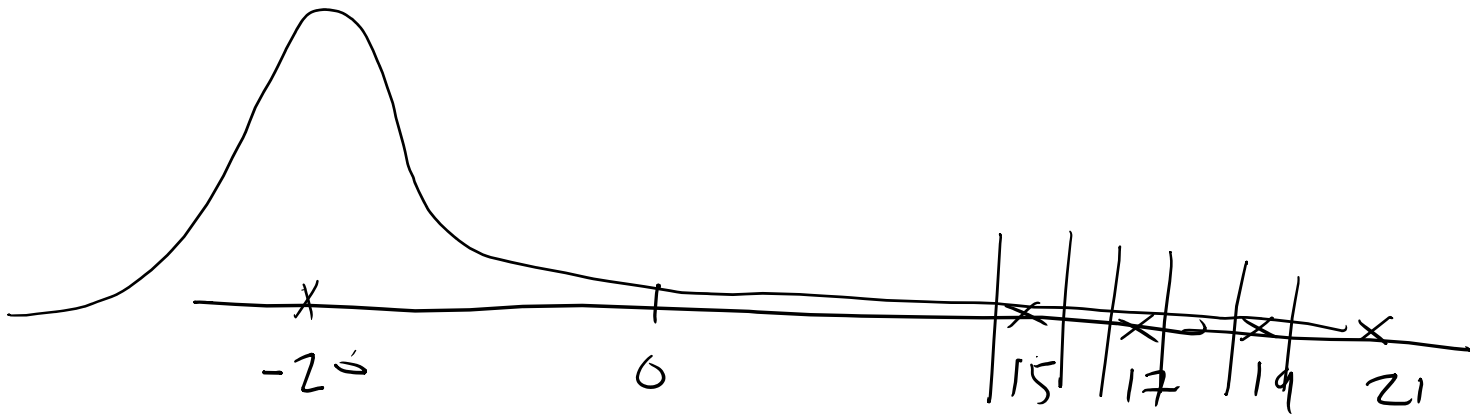
a standard normal random variable

$$\left(\frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}} \right)^2 \rightarrow \chi^2 \text{ random variable}$$

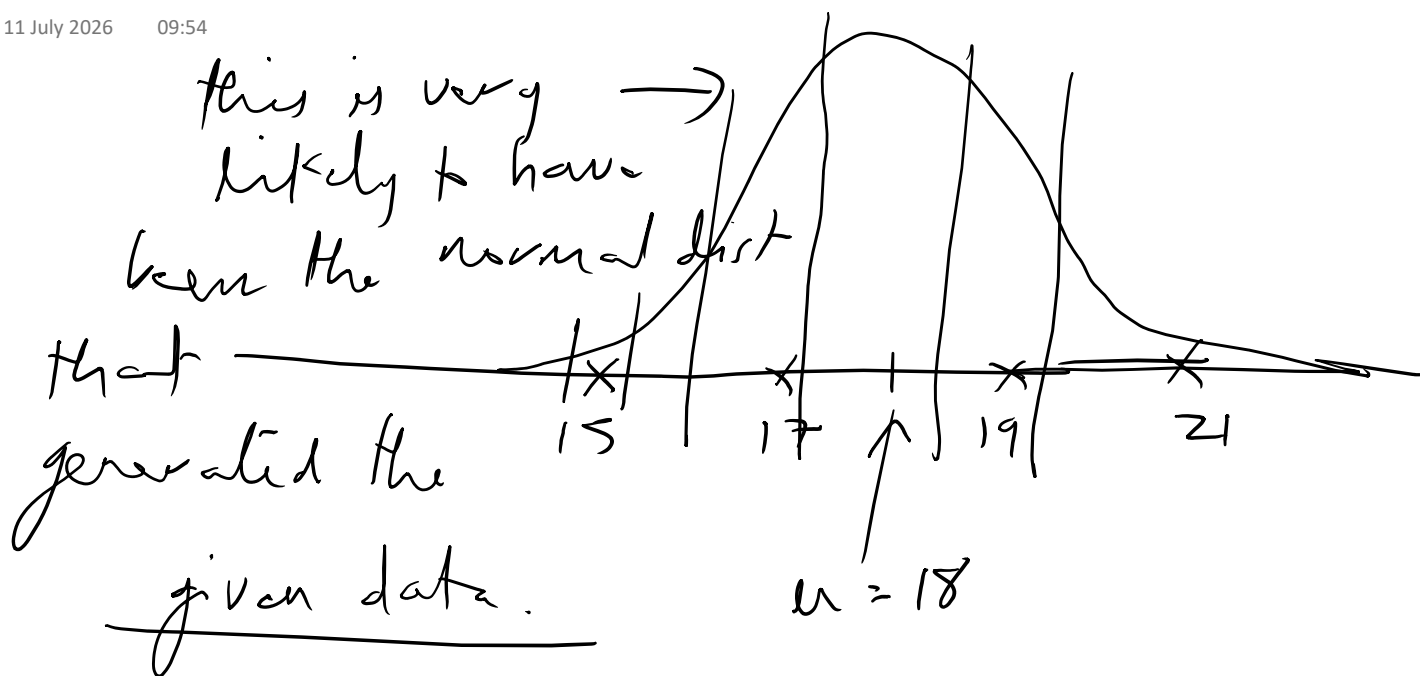


this data
has come out
of a
normal
distribution

What is the most likely normal
distribution that could have
produced this data?



Could a normal dist centered at $\mu = -20$
have produced this data?
↳ very, very unlikely



$$p^x (1-p)^{1-x}$$

$$P(X=1) = p^1 (1-p)^{1-1} = p \times (1-p)^0 = p$$

$$\underline{P(X=0)} = p^0 (1-p)^{1-0} = \underline{(1-p)}$$

$$P(X_1=1, X_2=0, X_3=1, X_4=X_5=\dots=X_9=0, X_{10}=1)$$

$$P(X_1=1) P(X_2=0) P(X_3=1) P(X_4=0) P(X_5=0) \\ P(X_{10}=1)$$

$$p (1-p) p (1-p)(1-p)$$

p occurs in this product as many times as you see a 1 in the

string
 $(1-p)$ occurs in the remaining places

$$\Rightarrow p^3 (1-p)^7$$

We need to find a value of p that maximizes the probability of observing the given string

$p^3(1-p)^7$ is the probability of observing the given string if $P(X_i = 1) = p$

$$\max_p p^3(1-p)^7$$

$\max_p p^3(1-p)^7$ is equivalent to
 $\max_p \log [p^3(1-p)^7]$ the same p maximizes $p^3(1-p)^7$ and

$$\log p^3(1-p)^7 = 3\log p + 7\log(1-p)$$

$$f(p) = 3 \log p + 7 \log (1-p)$$

$$\frac{df}{dp} = 0 \Rightarrow \frac{3}{p} + \frac{7(-1)}{(1-p)} = 0$$

$$\text{Since } \frac{d}{dp} \log p = \frac{1}{p}$$

$$\frac{3}{p} = \frac{7}{(1-p)}$$

$$3 - 3p = 7p$$

$$\Rightarrow 10p = 3 \Rightarrow \boxed{p = \frac{3}{10}}$$